

In [2]: `%pip install opencv-python`

```
Collecting opencv-python
  Downloading opencv_python-5.0.0.93-cp37-abi3-win_amd64.whl.metadata (20 kB)
Requirement already satisfied: numpy>=2 in d:\ai\envs\pythonlearning\lib\site-packages (from opencv-python) (2.3.5)
Downloading opencv_python-5.0.0.93-cp37-abi3-win_amd64.whl (44.0 MB)
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----- 11.0/44.0 MB 17.2 MB/s eta 0:00:02
----- 20.4/44.0 MB 20.7 MB/s eta 0:00:02
----- 27.8/44.0 MB 23.1 MB/s eta 0:00:01
----- 34.3/44.0 MB 24.5 MB/s eta 0:00:01
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----- 43.8/44.0 MB 26.1 MB/s eta 0:00:01
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Installing collected packages: opencv-python
Successfully installed opencv-python-5.0.0.93
Note: you may need to restart the kernel to use updated packages.
```

In [3]: `import cv2
from PIL import Image
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import numpy as np`

In [9]: `# قراءة الصورة
img = cv2.imread('images/cameraman.tif')

عرض الصورة
plt.imshow(img)
plt.title("Loaded with OpenCV")
plt.axis('off')
plt.show()`

Loaded with OpenCV



```
In [10]: # فتح الصورة
img2 = Image.open('images/lena_gray_256.tif')

# عرض الصورة بتدرج رمادي
plt.imshow(img2, cmap=cm.Greys_r)
plt.title("Loaded with PIL")
plt.axis('off')
plt.show()
```

Loaded with PIL



```
In [11]: # الحفظ باستخدام OpenCV
cv2.imwrite('new_image.jpg', img)

# الحفظ باستخدام PIL
img2.save('new_image2.jpg')
```

```
In [12]: # بشكل تلقائي NumPy تكون مصفوفة OpenCV صورة
print("OpenCV Image Shape:", img.shape)
print("OpenCV Image Data Type:", img.dtype)
print("OpenCV Image Array:\n", img)

print("-" * 50)

# NumPy array يلزم تحويلها أولاً إلى PIL صورة
img_array = np.array(img2)
print("PIL Image Shape:", img_array.shape)
print("PIL Image Data Type:", img_array.dtype)
print("PIL Image Array:\n", img_array)
```

OpenCV Image Shape: (256, 256, 3)

OpenCV Image Data Type: uint8

OpenCV Image Array:

```
[[[156 156 156]
  [159 159 159]
  [158 158 158]
  ...
  [151 151 151]
  [152 152 152]
  [152 152 152]]]
```

```
[[[160 160 160]
  [154 154 154]
  [157 157 157]
  ...
  [154 154 154]
  [155 155 155]
  [153 153 153]]]
```

```
[[[156 156 156]
  [159 159 159]
  [158 158 158]
  ...
  [151 151 151]
  [152 152 152]
  [152 152 152]]]
```

...

```
[[[114 114 114]
  [132 132 132]
  [123 123 123]
  ...
  [135 135 135]
  [137 137 137]
  [114 114 114]]]
```

```
[[[121 121 121]
  [126 126 126]
  [130 130 130]
  ...
  [133 133 133]
  [130 130 130]
  [113 113 113]]]
```

```
[[[121 121 121]
  [126 126 126]
  [130 130 130]
  ...
  [133 133 133]
  [130 130 130]
  [113 113 113]]]]
```

PIL Image Shape: (256, 256)

PIL Image Data Type: uint8

PIL Image Array:

```
[[162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 [162 162 162 ... 168 171 155]
 ...
 [ 48  57  54 ...  75  97  88]
 [ 46  50  52 ...  89 100  98]
 [ 44  55  54 ...  97 100 105]]
```

```
In [13]: # معرفة أقل وأعلى قيمة للبكسل
print("Min pixel value:", img_array.min())
print("Max pixel value:", img_array.max())

# حساب متوسط قيم البكسلات
print("Mean pixel value:", img_array.mean())

# أخذ جزء مقتطع من الصورة (Slicing/Cropping)
crop = img_array[50:150, 50:150]
plt.imshow(crop, cmap='gray')
plt.title("Cropped Region")
plt.axis('off')
plt.show()
```

Min pixel value: 26

Max pixel value: 242

Mean pixel value: 124.00845336914062

Cropped Region



In []: